UNIT - IV
Limit State Design of Columns & Footings

Design of Reinforced Concrete Structures (23A01502) - handwritten study notes

Syllabus Topics

- **Columns**: types of columns, design of short rectangular and circular columns for axial load, uniaxial bending and biaxial bending.
- **Footings**: wall footing, axially and eccentrically loaded rectangular pad and sloped footings, combined rectangular footing for two columns only.
- Code used: **IS 456 : 2000** (columns - clause 25, 26, 39; footings - clause 34) and **SP-16** design charts.

Exam tip: Unit IV always carries one column question (axial / uniaxial with SP-16 chart) and one footing question (isolated or combined). Learn the steps + one worked example of each.

1. Columns - Introduction

Definition

- A **column** is a vertical compression member whose effective length is more than **3 times** its least lateral dimension (otherwise it is a pedestal).
- It transfers the load of beams / slabs above to the foundation. Failure of a column is **sudden and brittle**, so a higher factor of safety is used.
- Material safety factors: concrete 1.5, steel 1.15; load factor 1.5 (DL + LL).

Classification of Columns

(a) Based on reinforcement and shape

- **Tied column**: longitudinal bars + lateral ties (rectangular, square, L-shaped). Most common.

- **Spiral (helically reinforced) column**: longitudinal bars + continuous helix, usually circular. More ductile and tough, takes 5% extra load.
- **Composite column**: structural steel section (or pipe) encased in or filled with concrete.

(b) Based on slenderness ratio

- **Short column**: $l_{ex}/D < 12$ and $l_{ey}/b < 12$ - fails by crushing of concrete / yielding of steel (material failure).
- **Long (slender) column**: ratio >= 12 - fails by **buckling**; additional moments M_a are to be added (clause 39.7).

(c) Based on loading

- **Axially loaded column** (theoretical, eccentricity only minimum e_{min}).
- **Uniaxial bending**: axial load + moment about one axis.
- **Biaxial bending**: axial load + moments about both axes (corner columns).

Effective Length (Table 28, IS 456)

- Effective length $l_e = k \times l$ (unsupported length). Depends on end conditions:

End condition	l_e / l
Fixed at both ends	0.65
Fixed one end, hinged other	0.80
Hinged both ends	1.00
Fixed one end, free other (cantilever)	2.00

- **Slenderness ratio** = l_e / D (D = depth in the direction considered). If $l_e/D < 12$, the column is short.

Minimum Eccentricity (clause 25.4)

$$e_{min} = l/500 + D/30 >= 20 \text{ mm}$$

- l = unsupported length of the column, D = lateral dimension in the direction under consideration.

- Every column must be designed for at least this eccentricity, in each direction.
- If e_{min} <= 0.05 D, the column can be designed for **axial load only** with the formula below.

For biaxial design, e_{min} is checked **separately** about each axis; but it is not applied simultaneously.

2. Code Requirements for Columns

Longitudinal Steel (clause 26.5.3.1)

- Minimum steel = **0.8%** of gross area A_g; maximum = **6%** (practically **4%** to ease concreting, lap zones need 6% max).
- Minimum **4 bars** in rectangular columns and **6 bars** in circular columns.
- Minimum bar diameter = **12 mm**.
- Cover to column bars: **40 mm** or the bar dia, whichever more (nominal cover 40 mm, 25 mm for small columns <= 200 mm and bars <= 12 mm).

Transverse Reinforcement (clause 26.5.3.2)

- **Lateral ties**: dia >= **1/4 of largest longitudinal bar** and not less than **6 mm**.
- **Pitch of ties** = least of:
  - least lateral dimension of column
  - 16 x dia of smallest longitudinal bar
  - 300 mm

Helical Reinforcement (clause 26.5.3.2 d)

- Pitch <= **75 mm** and <= core dia / 6; pitch >= **25 mm** and >= 3 x dia of helix.

3. Axially Loaded Short Columns (clause 39.3)

$$P_u = 0.4\, f_{ck}\, A_c + 0.67\, f_y\, A_{sc}$$

- P_u = factored axial load, A_c = net concrete area = $A_g - A_{sc}$, A_{sc} = area of longitudinal steel.
- The 0.4 and 0.67 factors already include the **minimum eccentricity** effect (0.45 f_{ck} and 0.75 f_y are the pure axial strengths, reduced by about 10% for eccentricity).
- **Helically reinforced** short columns: take **1.05 times** the strength:

$$P_u = 1.05\,(\,0.4\, f_{ck}\, A_c + 0.67\, f_y\, A_{sc}\,)$$

Design Steps - Axially Loaded Column

Factored load P_u → Assume p (1 to 2%) → Find A_g → Fix b and D

Find A_{sc} (→ = 0.8%) → Ties / detailing

Worked Example 1 - Square tied column

Q. Design a short square column to carry a **service load of 1000 kN**. Use M25 and Fe 415. Unsupported length 3.5 m.

- $P_u = 1.5 \times 1000 =$ **1500 kN**.
- Assume p = 1.5%: $1500 \times 10^3 = A_g\,[\,0.4 \times 25 \times 0.985 + 0.67 \times 415 \times 0.015\,] = A_g\,(9.85 + 4.17) = 14.02\, A_g$
- $A_g = 107{,}000\ mm^2 \longrightarrow$ provide **350 x 350 mm** ($A_g = 122{,}500\ mm^2$).
- Slenderness: $l_e/D = 3500 / 350 = 10 < 12$ so short. $e_{min} = 3500/500 + 350/30 = 18.7$ mm, i.e. less than 20 mm, so $e_{min} = 20$ mm. Since $e_{min} > 0.05\ D = 17.5$ mm, the 0.4 / 0.67 formula (which already allows for e_{min}) is used.
- $1500 \times 10^3 = 0.4 \times 25\,(122500 - A_{sc}) + 0.67 \times 415\, A_{sc} = 1{,}225{,}000 +$

268.05 A_{sc}

- A_{sc} = 275,000 / 268.05 = **1026 mm²** (min 0.8% = 980 mm², OK)
- Provide **4 - 20 mm dia** (1257 mm², p = 1.03%).
- Check: P_u = 0.4 x 25 x 121,243 + 0.67 x 415 x 1257 = 1212 + 350 = **1562 kN > 1500 kN, safe**.
- Ties: dia >= 20/4 = 5 mm and >= 6 mm ⟶ **8 mm** (use 8 mm as per practice). Pitch = least of (350, 16 x 20 = 320, 300) = **300 mm**.

Final: 350 x 350 column, 4 - 20 mm bars, 8 mm ties @ 300 mm c/c.

Worked Example 2 - Circular column with helical ties

Q. A circular column with helical reinforcement carries a factored load of **1800 kN**. M25, Fe 415. Design the column.

- Strength per unit: P_u/1.05 = 1800 / 1.05 = 1714 kN.
- Assume p = 1.5%: 1714 x 10^3 = 14.02 A_g ⟶ A_g = 122,270 mm².
- D = sqrt(4 A_g / pi) = sqrt(4 x 122270 / 3.1416) = 394.6 mm ⟶ provide **D = 400 mm**; A_g = 125,664 mm².
- 1,714,286 = 0.4 x 25 (125664 - A_{sc}) + 278.05 A_{sc} = 1,256,640 + 268.05 A_{sc}
- A_{sc} = 457,646 / 268.05 = **1707 mm²**, min 0.8% = 1005 mm² OK.
- Provide **6 - 20 mm** bars (1885 mm²).
- Helix: dia = 20/4 = 5 ⟶ use **8 mm**. Core dia = 400 - 2 x 40 = 320 mm; pitch <= core dia/6 = 53 mm; <= 75 mm; >= 25 mm and >= 3 x 8 = 24 mm. Provide **8 mm helix @ 50 mm pitch**.

Final: D = 400 mm, 6 - 20 mm bars, 8 mm helix at 50 mm pitch.

4. Columns under Uniaxial Bending (SP-16 charts)

- Columns in frames carry **P_u + M_u** about one axis. The exact analysis uses strain compatibility, stress block and the interaction diagram. SP-16 gives ready charts (Charts 27 to 62).

Interaction Diagram (P_u - M_u)

- Plot of axial capacity against moment capacity for a given section and steel.
- Points: pure axial (P_{uz} at zero moment), **balanced failure** (compression and tension failure together), pure bending (zero axial load).

Chart Parameters (non-dimensional)

$$P_u / (f_{ck}\, b\, D) \text{ and } M_u / (f_{ck}\, b\, D^2)$$

- d' = effective cover to the centre of the compression bars; ratio **d'/D** = 0.05, 0.10, 0.15, 0.20 gives separate charts.
- The chart gives **p/f_{ck}**, where $p = 100\, A_{sc} / (b\, D)$. So $p = (p/f_{ck}) \times f_{ck}$.

Design Procedure

- For long columns add the additional moment $M_a = P_u\, D\, (l_e/D)^2 / 2000$.

Worked Example 3 - Uniaxial bending

Q. A column 300 x 500 mm (D = 500 mm) is subjected to P_u = 1200 kN and M_u = 150 kN.m about the major axis. Effective length 3.5 m. M25, Fe 415, d' = 50 mm. Find A_{sc}.

- $e_{min} = 3500/500 + 500/30 = 7 + 16.7 = 23.7$ mm; $e_{actual} = M_u/P_u = 150/1.2 = 125$ mm > e_{min}, so design for M_u = 150.
- $P_u / (f_{ck}\, b\, D) = 1200 \times 10^3 / (25 \times 300 \times 500) =$ **0.32**
- $M_u / (f_{ck}\, b\, D^2) = 150 \times 10^6 / (25 \times 300 \times 500^2) =$ **0.08**
- $d'/D = 50/500 =$ **0.10** ⟶ use SP-16 chart for Fe 415, d'/D = 0.1 (steel on two faces, Chart 32).
- From the chart: p/f_{ck} ~ 0.04 (approximately - the exact value is read from your chart) ⟶ $p = 0.04 \times 25 =$ **1.0%**
- $A_{sc} = 1.0 \times 300 \times 500 / 100 =$ **1500 mm^2** ⟶ provide 4 - 22 mm (1520 mm^2).

5. Columns under Biaxial Bending (clause 39.6)

- Corner columns and columns with unsymmetrical beams carry **P_u**, **M_{ux}** **and** **M_{uy}** together. Exact analysis is complex, so IS 456 gives the **Bresler** **load** **contour** method.

Bresler's Formula

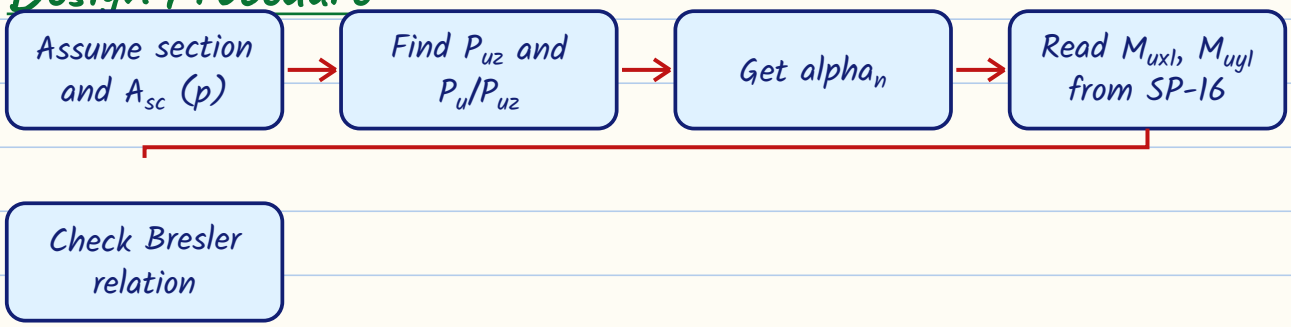
$$(M_{ux} / M_{uxl})^{alpha\ n} + (M_{uy} / M_{uyl})^{alpha\ n} <= 1.0$$

- M_{ux}, M_{uy} = factored moments about x and y axes (including e_{min} effects).
- M_{uxl}, M_{uyl} = **maximum** **uniaxial** **moment** **capacity** about x and y axes for the given P_u (read from SP-16 charts, with steel provided).
- $alpha_n$ depends on P_u/P_{uz}:

$$alpha_n = 1 + (P_u/P_{uz} - 0.2) / 0.6 \text{ for } 0.2 \text{ to } 0.8$$

- **P_{uz}** (squash load) = $0.45\ f_{ck}\ A_c + 0.75\ f_y\ A_{sc}$

Design Procedure



Worked Example 4 - Biaxial

Q. Check a 400 x 400 mm column, M25, Fe 415, with 1.8% steel (2880 mm^2, d'/D = 0.1) for P_u = 1000 kN, M_{ux} = 120 kN.m, M_{uy} = 80 kN.m.

- A_c = 160,000 - 2880 = 157,120 mm^2
- P_{uz} = 0.45 x 25 x 157,120 + 0.75 x 415 x 2880 = 1,767,600 + 896,400 = **2664 kN**
- P_u/P_{uz} = 1000/2664 = 0.375 ⟶ $alpha_n$ = 1 + (0.375 - 0.2)/0.6 = **1.29**
- $P_u/(f_{ck}\ b\ D)$ = 1000 x 10^3 / (25 x 400 x 400) = 0.25; p/f_{ck} = 1.8/25 =

0.072

- From SP-16 chart: $M_{ul}/(f_{ck}\ b\ D^2) \sim 0.119 \longrightarrow M_{uxl} = M_{uyl} \sim 0.119 \times 25 \times 400^3 =$ **190 kN.m** (chart reading, approximate)
- $(120/190)^{1.29} + (80/190)^{1.29} = 0.553 + 0.328 =$ **0.88 < 1.0 ⟶ safe**.

6. Footings - Introduction

- A **footing (foundation)** spreads the load of a column / wall over a large area of soil so that the **safe bearing capacity (SBC)** is not exceeded and settlement is within limits.

Design Principles (clause 34)

- **Plan area** from **service (working) loads** and SBC: A = (W + weight of footing, about 10% of W) / SBC.
- **Depth, reinforcement** from **factored loads** (limit state), with soil pressure $p_u = 1.5 \times p$ (upward, net of footing weight).
- Soil pressure for structural design is **net upward pressure** (self-weight of footing does not cause bending).

Critical Sections

Check	Critical section	Notes
Bending moment	At the **face of column** (wall)	Cantilever projection
One-way (beam) shear	At **distance d** from column face	$tau_v <= tau_c$ (Table 19)
Two-way (punching) shear	At **d/2** from column face all round	$tau_v <= k_s\ tau_c'$, $tau_c' = 0.25\ sqrt(f_{ck})$
Development length	At column face and at the changes of section	$L_d = 0.87\ f_y\ phi / (4\ tau_{bd})$

- $k_s = 0.5 + beta_c <= 1$, where $beta_c$ = short side / long side of the column.

7. Design of Wall Footing

- Main steel is **across** the wall (transverse); **distribution** steel is **along** the wall (0.12% of area).

Procedure

Load per metre W → Width $B = 1.1\,W$ / SBC → $p_u = 1.5\,W / B$ → Projection a

$M_u = p_u\,a^2/2$ → d from shear and moment → A_{st}, distribution steel

Worked Example 5

Q. Design a strip footing for a 230 mm thick wall carrying **150 kN/m** (service) on soil of **SBC 150 kN/m²**. Use M20 and Fe 415.

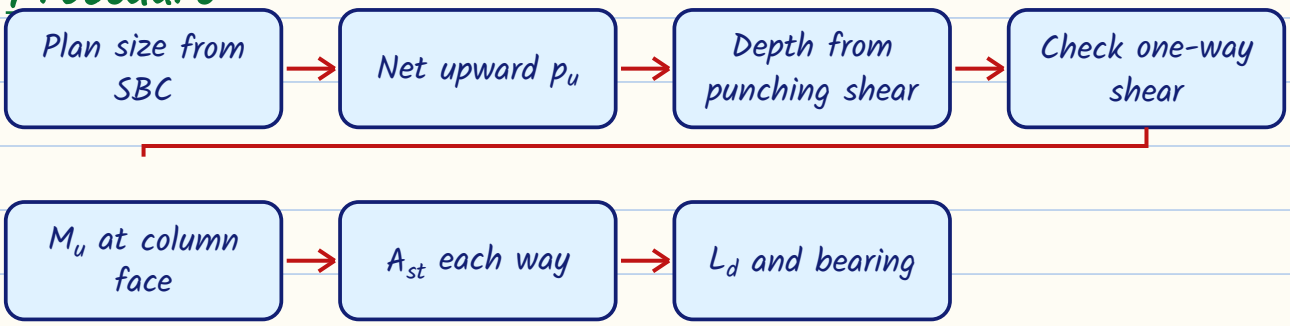
- Width: $B = 1.1 \times 150 / 150 =$ **1.1 m**.
- Net factored pressure: $p_u = 1.5 \times 150 / 1.1 =$ **204.5 kN/m²**.
- Projection $a = (1.1 - 0.23) / 2 =$ **0.435 m**.
- Mu at face of wall $= p_u\,a^2 / 2 = 204.5 \times 0.435^2 / 2 =$ **19.35 kN.m per m**.
- Depth from moment: $d^2 = 19.35 \times 10^6 / (0.138 \times 20 \times 1000) = 7011 \longrightarrow d = 84$ mm (small). **Shear** and min edge depth govern. Try $D = 250$ mm, $d = 250 - 50 - 5 =$ **195 mm**.
- One-way shear at d from face: $V_u = 204.5\,(0.435 - 0.195) = 49.1$ kN; $\tau_v = 49.1 \times 10^3 / (1000 \times 195) =$ **0.25 N/mm²**.
- τ_c for $p_t \sim 0.15\%$ (M20) ~ 0.28 N/mm² > 0.25. **Safe**.
- A_{st}: $19.35 \times 10^6 = 0.87 \times 415 \times A_{st} \times 195\,(1 - \ldots) \longrightarrow A_{st} \sim 289$ mm²; minimum steel $= 0.12\% \times 1000 \times 250 =$ **300 mm²**.
- Provide **8 mm @ 150 mm c/c** (335 mm²) across the wall.
- Development length $L_d = 0.87 \times 415 \times 8 / (4 \times 1.2 \times 1.6) = 376$ mm; available $= 435 - 50 = 385$ mm $>= 376$ **OK**.
- Distribution steel $= 0.12\% \times 1000 \times 250 = 300$ mm² → **8 mm @ 160**

mm c/c along the wall.

> Final: B = 1.1 m, D = 250 mm, 8 mm @ 150 main (transverse), 8 mm @ 160 distribution.

8. Isolated Rectangular (Pad) Footing - Axial Load

Procedure



- **Plan size**: A = 1.1 P / SBC. Choose L/B equal to column D/b (so that projections are equal) - square for square columns.
- **Depth by punching shear** (critical perimeter at d/2): $b_o = 4 (c + d)$ for a square column of side c.
  - $V_u = p_u [B^2 - (c + d)^2]$; $\tau_v = V_u / (b_o d) \le k_s \tau_c'$
- **Bending moment**: $M_u = p_u B a^2 / 2$ where a = projection from the column face. Same in both directions for square footing.

Worked Example 6

Q. Design a square footing for a **400 x 400 mm** column carrying **1000 kN** service load. SBC = **200 kN/m²**. M25, Fe 415.

- Area = 1.1 x 1000 / 200 = 5.5 m^2 ⟶ **2.4 m x 2.4 m** (5.76 m^2).
- p_u = 1.5 x 1000 / 5.76 = **260.4 kN/m²** (net, upward).
- Projection a = (2.4 - 0.4)/2 = **1.0 m**.
- M_u = 260.4 x 2.4 x 1.0^2 / 2 = **312.5 kN.m**
- **Depth by punching shear** (try d = 490 mm): b_o = 4 (400 + 490) = 3560 mm
  - V_u = 260.4 [5.76 - 0.89^2] = 260.4 x 4.968 = 1294 kN

  - tau_v = 1294 x 10^3 / (3560 x 490) = **0.74** **N/mm²**
  - allowable = k_s x 0.25 sqrt(25) = 1 x 1.25 = 1.25 N/mm² > 0.74 **OK**
- Provide **D = 550 mm**, d = 550 - 50 - 10 = **490 mm**.
- **One-way shear** at d from face: V_u = 260.4 x 2.4 x (1.0 - 0.49) = 318.7 kN. tau_v = 318.7 x 10^3 / (2400 x 490) = **0.27** **N/mm²**.
- **Steel**: A_{st} = 312.5 x 10^6 / (0.87 x 415 x 0.95 x 490) = **1859** **mm²** ⟶ p_t = 0.158% ⟶ tau_c (M25) ~ 0.29 > 0.27 **Safe**.
- Minimum steel = 0.12% x 2400 x 550 = 1584 mm² < 1859 OK.
- Provide **12 mm @ 140 mm c/c** both ways (17 bars = 1923 mm²).
- L_d = 0.87 x 415 x 12 / (4 x 1.4 x 1.6) = 484 mm < available (1000 - 50 = 950 mm) **OK**.

Final: 2.4 x 2.4 m footing, D = 550 mm, 12 mm @ 140 c/c in both directions.

9. Eccentrically Loaded Footing

- Occurs when the column carries a **moment** M (or is not at the centroid of the base). Load P at eccentricity **e = M / P** from the centre line.
- The soil pressure is trapezoidal / triangular:

$$p = P/A\ (1 +/- 6e / L)$$

- p_{max} = P/(BL) (1 + 6e/L) and p_{min} = P/(BL) (1 - 6e/L), L = length in the direction of eccentricity.
- **No tension condition**: e <= L/6 (load within the **middle third**). Then p_{min} >= 0.

Procedure

Numerical illustration

- P = 800 kN, e = 0.3 m, footing 2.5 m x 2.0 m (L = 2.5 m). A = 5 m², P/A = 160 kN/m².
- 6e/L = 6 x 0.3 / 2.5 = 0.72 ; e = 0.3 < L/6 = 0.417 ⟶ no tension.

- p_{max} = 160 x 1.72 = **275.2 kN/m²**, p_{min} = 160 x 0.28 = **44.8 kN/m²**.
- Check p_{max} <= SBC (say 300 kN/m²) **OK**. Factored: $p_{u\ max}$ = 1.5 x 275.2 = 412.8 kN/m².

> Eccentric footing: e <= L/6 is the condition for no tension. p_{max} = (P/A)(1 + 6e/L) is the most asked formula.

10. Sloped (Stepped) Footing

- When the projection is large, the thickness needed at the column face is large while at the edge only a small depth is needed. A **sloped (tapered)** footing saves concrete.
- **Minimum edge thickness**: 150 mm (on soil); slope should be gentle (usually <= 35 degrees, so that the top surface can be cast without top formwork).

11. Combined Rectangular Footing for Two Columns

When used

- An **exterior column is on the property line** and cannot project outside.
- Combined footing is a **rectangle** if the two column loads are nearly equal or the outer edge is not restricted, **trapezoidal** if the loads are unequal and the length is restricted; **strap** if the columns are far apart.

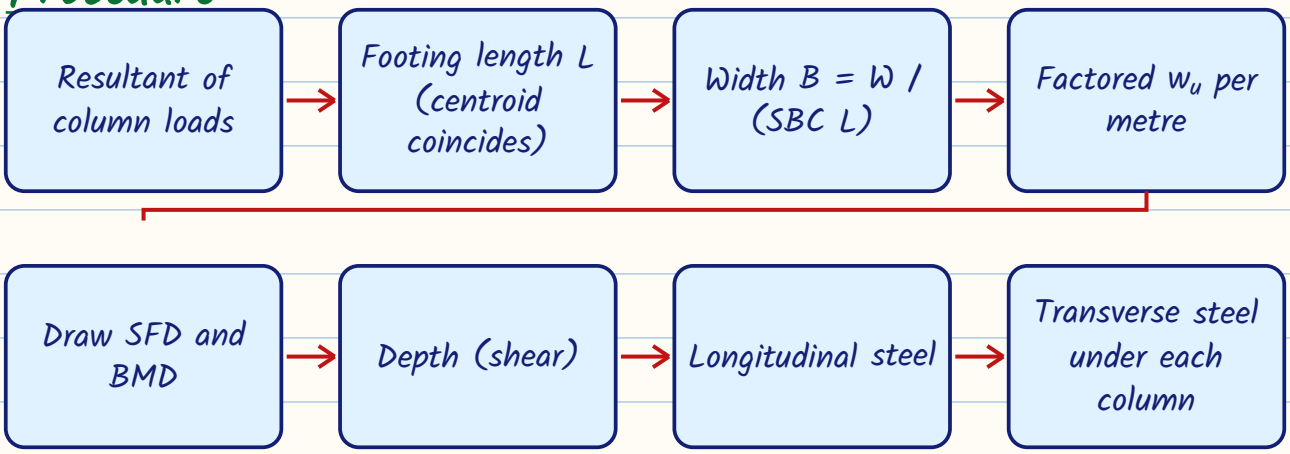
Basic Principle

- The **centroid of the footing area must coincide with the centroid (resultant) of the two column loads**, so that the soil pressure is **uniform**.

$$x_{bar} = P_2\ s\ /\ (P_1 + P_2)\ \text{(from column 1)}$$

- s = centre-to-centre spacing of columns; the footing length L = 2 (x_{bar} + a), where a = distance from the centre of column 1 to the nearer end.
- B = (P_1 + P_2 + 10%) / (SBC x L).

Procedure



- **Longitudinal direction**: footing acts as an **inverted beam** (upward pressure, loads from columns). Maximum **hogging** moment occurs between the columns where the **shear force = 0** ⟶ **top steel**.
- **Transverse direction**: each column acts with a **transverse beam strip** of width (c + d) on each side; the strip acts like a cantilever under the column ⟶ **bottom steel**.
- Check **one-way shear** at d from face of each column and **punching shear** around each column.

Worked Example 7

Q. Design a combined footing for two columns 400 x 400 mm; C1 carries 800 kN and C2 carries 1200 kN (service). Spacing 4 m c/c. SBC = 200 kN/m^2. M25, Fe 415.

- Total service load = 2000 kN; with 10% = 2200 kN. Area = 2200/200 = **11 m^2**.
- Resultant from C1: x_{bar} = 1200 x 4 / 2000 = **2.4 m**.
- Let the left end be 0.4 m from the centre of C1: L = 2 (2.4 + 0.4) = **5.6 m** (right end projects 1.2 m beyond C2).
- B = 11 / 5.6 = 1.96 ⟶ **B = 2.0 m** (area 11.2 m^2).
- Factored loads: P_{u1} = 1200 kN, P_{u2} = 1800 kN, total 3000 kN.
- Upward pressure w_u = 3000 / 5.6 = **535.7 kN/m** (= 267.9 kN/m^2 x 2 m).

Shear force and bending moment (measured from left end)

Section	Shear force (kN)
Left end (x = 0)	0
Just left of C1 (x = 0.4)	535.7 x 0.4 = 214.3
Just right of C1	214.3 - 1200 = -985.7
Just left of C2 (x = 4.4)	535.7 x 4.4 - 1200 = 1157
Just right of C2	1157 - 1800 = -643
Right end (x = 5.6)	-643 + 535.7 x 1.2 = 0

- Zero shear at x = 0.4 + 985.7/535.7 = 0.4 + 1.84 = **2.24 m** from the left end.
- **Maximum hogging moment** M_u = 535.7 x $2.24^2/2$ - 1200 x 1.84 = 1344 - 2208 = **-864 kN.m**.
- **Cantilever moment** at the face of C2 (right side, 1.0 m): 535.7 x $1.0^2/2$ = 268 kN.m (sagging).

Depth - check shear: try D = 1000 mm, d = 940 mm.

- Critical section at d from face of C2 (face at x = 4.2, section at x = 3.26): V_u = 535.7 x 3.26 - 1200 = **546 kN**.
- tau_v = 546 x 10^3 / (2000 x 940) = **0.29 N/mm²**.
- **Longitudinal steel** (top): A_{st} = 864 x 10^6 / (0.87 x 415 x 0.9 x 940) = 2829 mm² ⟶ provide **10 - 20 mm** bars (3142 mm²), p_t = 0.167%, tau_c ~ 0.30 N/mm² > 0.29 **Safe** (shear).
- Minimum steel = 0.12% x 2000 x 1000 = 2400 mm² OK. Bottom steel under the cantilever: provide minimum steel (2400 mm²) ⟶ 16 mm @ 150 mm.
- **Punching shear** at C2: b_o = 4 (400 + 940) = 5360 mm; V_u = 1800 - 267.9 x 1.34^2 = 1319 kN; tau_v = 1319 x 10^3/(5360 x 940) = 0.26 N/mm² < 1.25 **OK**.

Final: L = 5.6 m, B = 2.0 m, D = 1000 mm; top steel 10 - 20 mm longitudinal; bottom min steel + transverse bars under each column.

12. Quick Revision

Formulas to remember

- e_{min} = l/500 + D/30 >= 20 mm
- Axial short column: $P_u = 0.4 f_{ck} A_c + 0.67 f_y A_{sc}$; helical: 1.05 x
- $P_{uz} = 0.45 f_{ck} A_c + 0.75 f_y A_{sc}$; $alpha_n$: 1 (P_u/P_{uz} <= 0.2) to 2 (>= 0.8)
- Bresler: $(M_{ux}/M_{uxl})^{alpha\ n} + (M_{uy}/M_{uyl})^{alpha\ n}$ <= 1
- Eccentric footing: p = (P/A)(1 +/- 6e/L), e <= L/6

Important Questions

- 1. Classify columns. Explain short and long columns with slenderness limits.
- 2. State the code requirements for longitudinal and transverse reinforcement of columns.
- 3. Design a short square / circular column for the given axial load (tied / helical).
- 4. Explain the interaction diagram and design a rectangular column for axial load and uniaxial moment using SP-16.
- 5. Explain the Bresler load contour method for biaxial bending with an example.
- 6. Design a wall footing for a given load per metre and SBC.
- 7. Design an isolated square / rectangular footing for a column carrying axial load.
- 8. Derive the soil pressure equation for an eccentrically loaded footing. What is the condition for no tension?